

Topic 12: Method of Undetermined Coefficients & The Annihilator Method

Professor Leonard Master Series

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Educated Guess & The Resonance Clash

Professor Leonard explains the non-homogeneous equation $ay'' + by' + cy = g(x)$:

"The complete general solution is always a partnership: $y(x) = y_c(x) + y_p(x)$. y_c is the complementary solution satisfying the homogeneous part $= 0$. y_p is a particular solution engineered to match the forcing driving function $g(x)$. But here is the ultimate exam trap: If your guess for y_p duplicates ANY term already living in y_c , the differential operator will wipe it out to zero! When a clash occurs, you MUST multiply your guess by x (or x^2) until the collision is broken!"

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 12.1: Polynomial & Exponential Driving Terms with Duplication Clash

Statement: Solve $y'' - 3y' - 4y = 2e^{-x} + 4x^2$.

Step-by-Step Whiteboard Solution:

- **Step 1: Solve Homogeneous Part** $y_c(x)$:

$$r^2 - 3r - 4 = 0 \implies (r - 4)(r + 1) = 0 \implies r = 4, -1$$

$$y_c(x) = c_1 e^{4x} + c_2 e^{-x}$$

- **Step 2: Inspect Driving Terms:** $g(x) = 2e^{-x} + 4x^2$.
 - Notice e^{-x} is **already in** $y_c(x)$! A regular guess Ae^{-x} will give 0!
 - Therefore, multiply by x : Guess $y_{p1} = Axe^{-x}$.
 - For $4x^2$: Guess complete 2nd-degree polynomial: $y_{p2} = Bx^2 + Cx + D$.
 - Combined Guess:

$$y_p(x) = Axe^{-x} + Bx^2 + Cx + D$$

- **Step 3: Differentiate** $y_{p1} = Axe^{-x}$:

$$y'_{p1} = A(e^{-x} - xe^{-x})$$

$$y_{p1}'' = A(-e^{-x} - e^{-x} + xe^{-x}) = A(xe^{-x} - 2e^{-x})$$

Substitute into LHS:

$$y'' - 3y' - 4y = Ae^{-x}[(x-2) - 3(1-x) - 4x] = Ae^{-x}[x-2-3+3x-4x] = -5Ae^{-x}$$

Set equal to $2e^{-x} \implies -5A = 2 \implies A = -\frac{2}{5}$.

- **Step 4: Differentiate** $y_{p2} = Bx^2 + Cx + D$:

$$y_{p2}' = 2Bx + C, \quad y_{p2}'' = 2B$$

Substitute:

$$\begin{aligned} 2B - 3(2Bx + C) - 4(Bx^2 + Cx + D) &= 4x^2 \\ -4Bx^2 + (-6B - 4C)x + (2B - 3C - 4D) &= 4x^2 + 0x + 0 \end{aligned}$$

- **Step 5: Equate Coefficients:**

- x^2 : $-4B = 4 \implies B = -1$.

- x : $-6(-1) - 4C = 0 \implies 6 - 4C = 0 \implies C = \frac{3}{2}$.

- Constant: $2(-1) - 3(3/2) - 4D = 0 \implies -2 - \frac{9}{2} - 4D = 0 \implies -\frac{13}{2} = 4D \implies D = -\frac{13}{8}$.

- **Step 6: Assemble Full General Solution:**

$$y(x) = c_1 e^{4x} + c_2 e^{-x} - \frac{2}{5} x e^{-x} - x^2 + \frac{3}{2} x - \frac{13}{8}$$

Problem 12.2: Pure Harmonic Resonance Clash

Statement: Solve $y'' + 4y = 3 \sin 2x$.

Step-by-Step Whiteboard Solution:

- **Step 1: Find** $y_c(x)$:

$$r^2 + 4 = 0 \implies r = \pm 2i \implies y_c(x) = c_1 \cos 2x + c_2 \sin 2x$$

- **Step 2: Resonance Detection:** $g(x) = 3 \sin 2x$ has natural frequency $\omega = 2$, which perfectly matches the natural frequency of the system! Standard guess $A \cos 2x + B \sin 2x$ is in y_c and produces 0! **Multiply by** x :

$$y_p(x) = x(A \cos 2x + B \sin 2x) = Ax \cos 2x + Bx \sin 2x$$

- **Step 3: Differentiate and Substitute:**

$$y_p' = A(\cos 2x - 2x \sin 2x) + B(\sin 2x + 2x \cos 2x)$$

$$y_p'' = -4A \sin 2x - 4Ax \cos 2x + 4B \cos 2x - 4Bx \sin 2x$$

$$y_p'' + 4y_p = (-4A \sin 2x + 4B \cos 2x) = 3 \sin 2x$$

- **Step 4: Match Coefficients:**
 - $\sin 2x: -4A = 3 \implies A = -\frac{3}{4}.$
 - $\cos 2x: 4B = 0 \implies B = 0.$
- **Step 5: Particular and General Solution:**

$$y_p(x) = -\frac{3}{4}x \cos 2x$$

$$y(x) = c_1 \cos 2x + c_2 \sin 2x - \frac{3}{4}x \cos 2x$$

Problem 12.3: Repeated Root Double Clash (x^2 Factor)

Statement: Solve $y'' - 2y' + y = 4e^x$.

Step-by-Step Whiteboard Solution:

- **Step 1: Complementary Solution:**

$$r^2 - 2r + 1 = 0 \implies (r - 1)^2 = 0 \implies y_c(x) = c_1 e^x + c_2 x e^x$$

- **Step 2: Double Clash:**
 - Guess Ae^x ? Clashes with $c_1 e^x$!
 - Guess Axe^x ? Clashes with $c_2 x e^x$!
 - Must multiply by x AGAIN:

$$y_p(x) = Ax^2 e^x$$

- **Step 3: Substitute and Solve:**

$$y_p' = A(2xe^x + x^2 e^x)$$

$$y_p'' = A(2e^x + 4xe^x + x^2 e^x)$$

$$y_p'' - 2y_p' + y_p = Ae^x[(2 + 4x + x^2) - 2(2x + x^2) + x^2] = 2Ae^x$$

$$\text{Set equal to } 4e^x \implies 2A = 4 \implies A = 2.$$

- **Step 4: Final Solution:**

$$y(x) = c_1 e^x + c_2 x e^x + 2x^2 e^x$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Applying Initial Conditions to y_c alone:** You CANNOT solve for c_1, c_2 until you have added y_p ! Initial conditions apply to the TOTAL solution $y = y_c + y_p$.
- **Trap 2: Truncating Polynomial Guesses:** If $g(x) = 4x^2$, your guess MUST be $Ax^2 + Bx + C$, including the lower-order linear and constant terms!